

Yilang Liu

New Haven, CT (open to relocation) ✉ yilang.liu@yale.edu 🌐 yilangliu.github.io
🐙 github.com/YilangLiu 🔗 linkedin.com/in/yilangliu 📄 Google Scholar

SUMMARY

Robotics and machine learning researcher specializing in reinforcement learning, differentiable simulation, and vision-language-action (VLA) policies deployed on real robots — factory-validated bimanual mobile manipulation and sim-to-real legged locomotion (Unitree Go2). Publications at NeurIPS 2025 (**Spotlight**), ICRA 2026, and RSS 2026. Yale Ph.D. expected May 2027; seeking Robotics/ML Engineer and Research Scientist roles (open to final-year internships).

EDUCATION

Yale University	New Haven, CT
<i>Ph.D. in Mechanical Engineering, focus on robotics & robot learning (M.S./M.Phil. en route)</i>	<i>Expected May 2027</i>
Carnegie Mellon University	Pittsburgh, PA
<i>M.S. in Mechanical Engineering (Research Program)</i>	<i>May 2022</i>
University of Cincinnati & Chongqing University (Joint Co-op Institute)	Cincinnati, OH / Chongqing, China
<i>B.S. in Mechanical Engineering (cum laude); B.E. in Mechanical Design, Manufacture and Automation</i>	<i>Apr 2020</i>

SKILLS

Languages: Python, C++.

ML & RL: vision-language-action model fine-tuning & deployment (GR00T N1.6, pi-0.5), PyTorch, reinforcement learning (PPO, policy gradients, visual RL), imitation learning (ACT), differentiable simulation, sim-to-real transfer.

Robotics & Control: ROS, ROS 2, motion planning (RRT), sampling-based MPC, state estimation (EKF), real-time torque-level control, teleoperation; Unitree A1/Go2 quadrupeds.

Simulation & Tools: MuJoCo, Isaac Sim, Isaac Gym, Genesis, Brax, PyTorch3D, LeRobot, Weights & Biases, Linux, Git.

PROFESSIONAL EXPERIENCE

Dexmate Inc.	Santa Clara, CA
<i>Robotics Intern</i>	<i>Oct 2025 – Jan 2026</i>

- Post-trained and deployed the GR00T N1.6 VLA on [Dexmate Vega](#), a dual-arm mobile manipulator, for an Amazon-commissioned **2-minute** bimanual box-unfolding task: collected ~80 teleoperated demonstrations, converted to the LeRobot format, and full fine-tuned the model after zero-shot deployment failed — **80% success over 10 trials**.
- Fine-tuned and deployed the pi-0.5 ($\pi_{0.5}$) VLA for a bimanual client task of placing a condenser on a conveyor; smoothed action-chunk transitions with Real-Time Action Chunking (RTC) at **~20 Hz**; **90% success over 20 factory trials**.
- Distributed both VLA fine-tuning runs across two NVIDIA RTX Pro 6000 GPUs with torchrun.
- Deployed an Action Chunking with Transformers (ACT) imitation-learning policy trained on a bimanual teleoperated demonstration dataset, achieving reliable vision-based bottle grasping and placement.
- Trained a PPO grasping policy for a 6-DoF dexterous hand in Genesis and Isaac Sim, guided by RRT motion planning.

RESEARCH EXPERIENCE

Graduate Research Assistant, Yale University	New Haven, CT
<i>Intelligent Autonomy Lab; Advisor: Prof. Ian Abraham</i>	<i>2022 – Present</i>

Efficient on-policy visual RL via decoupled policy gradients (SDPG) CVPR 2026 Workshop [\[paper\]](#) [\[code\]](#)

- Developed SDPG, a visual RL method estimating policy gradients from random perturbations of trajectory rollouts, without differentiating through rendering; trains visuomotor policies **in hours on a single RTX 4080 GPU**.
- Outperformed DrQ-v2, TD-MPC2, DreamerV3, and PPO on visual MuJoCo benchmarks in training time, memory usage, and final reward.
- Released an open-source suite of egocentric visual RL benchmarks spanning dexterous manipulation and locomotion (🐙); demonstrated sim-to-real transfer on a Unitree Go2 quadruped.

Parallel differentiable simulation for visual policy learning (D.Va) NeurIPS 2025 (**Spotlight**) [\[paper\]](#) [\[code\]](#)

- Co-developed D.Va, cutting visual-policy training time by **10x** by decoupling image observations from the differentiable-simulation computation graph (no backpropagation through rendering); trained a humanoid running policy from pixels in **4 hours on a single GPU**.
- Achieved **5x higher reward** than model-free and model-based baselines on humanoid running; built a PyTorch3D differentiable-rendering baseline, showing that decoupling yields smoother first-order gradients than backpropagating through rendering.

Sample-based hybrid mode control

ICRA 2026 (earlier workshop version at RSS 2025) [\[paper\]](#)

- Designed a sampling-based control strategy for high-dimensional, long-horizon tasks grounded in hybrid control theory, with proven performance guarantees on mode-sequence optimization.
- Enabled contact-rich in-hand manipulation and complex mode-switching behaviors, achieving **85%** and **41%** higher task reward than baselines in manipulation and humanoid locomotion, respectively.
- Deployed **real-time torque-level** sampling-based predictive control on a Unitree Go2 — including an EKF-based proprioceptive state estimator — achieving standing-to-handstand transitions on hardware.

Ergodic coverage on generalized motion fields

RSS 2026 [\[paper\]](#)

- Developed a flow-adaptive ergodic trajectory-optimization framework using maximum mean discrepancy (MMD) for robotic exploration in time-varying flow fields, with coverage guarantees on evolving domains (8-author, multi-institution collaboration).
- Implemented a sampling-based trajectory optimizer for non-convex, flow-constrained environments; validated on Crazyflie drones and a Unitree Go2 in simulation and hardware, with up to **42% higher exploration performance** than information-theoretic baselines.

SELECTED PUBLICATIONS

H. You, **Y. Liu**, I. Abraham. “[Accelerating Visual-Policy Learning through Parallel Differentiable Simulation](#)”. NeurIPS 2025 (**Spotlight**).

Y. Liu, H. You, I. Abraham. “[Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes](#)”. ICRA 2026.

C. Hughes, **Y. Liu**, J. Engdahl, H. Warren, D. Lee, F. Ramos, T. Miles, I. Abraham. “[Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields](#)”. RSS 2026.

H. You*, **Y. Liu***, I. Abraham. “[Efficient On-policy Visual-RL via Stochastic Decoupled Policy Gradient](#)”. CVPR 2026 Workshop (*equal contribution).

Y. Liu, A. Barati Farimani. “[An Energy-Saving Snake Locomotion Pattern Learned in a Physically Constrained Environment With Online Model-Based Policy Gradient Method](#)”. ASME J. Mechanisms and Robotics, 2023.

Full list: [Google Scholar](#).

TEACHING & SERVICE

Teaching: Teaching Fellow, Yale ENAS 773 (Introduction to Robotics, Control, and Learning); Teaching Assistant, CMU 18-461 (Introduction to Machine Learning for Engineers).

Reviewer: ICRA (2022–2026), IROS (2022–2026), RSS (2025), ECCV (2024); IEEE Robotics and Automation Letters (RA-L).